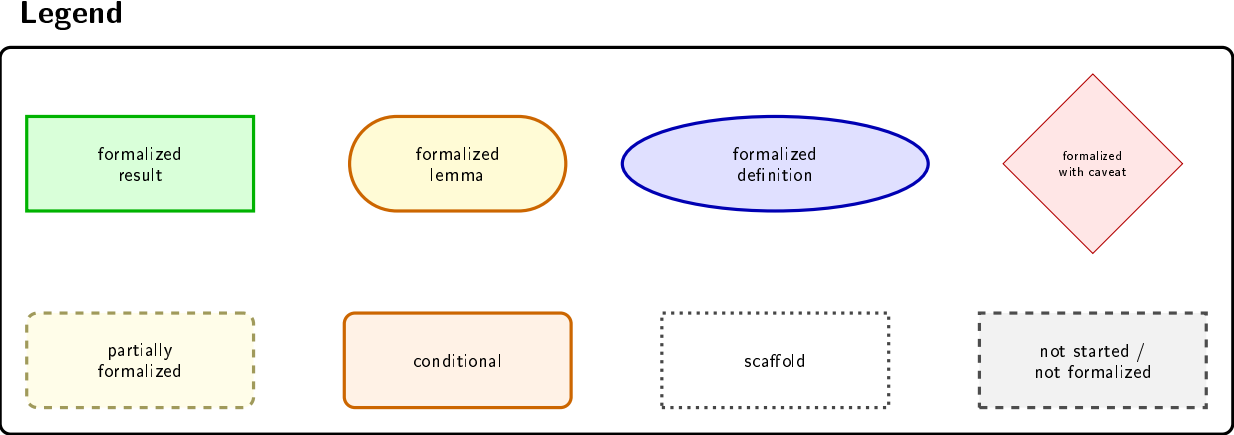




Sections 2–3
slots, click rates, values,
GSP and VCG pay-
ments, truthfulness and
equilibrium predicates

Definition 4
locally envy-
free equilibrium

Remark 1
same-bid GSP per-
click payment weakly
dominates VCG by a
generic ranked tail-
payment induction

Remark 2
VCG-style position
mechanisms are
dominant-strategy
truthful from the welfare-
maximization/externality-
tax certificate

Remark 3
truth-telling is not
dominant under GSP,
by the paper’s three-
bidder/two-slot deviation

Sections 2.2–2.3
concrete GSP wit-
ness; running example
Nash, LEF, payments,
and VCG-minimality

Lemma 5
 $LEF \Rightarrow$ formal stable as-
signment; caveat: stronger
no-rematch envy predicate

Lemma 6
stable assignment $\Rightarrow$
slot envy-free; caveat:
VCG-payment lat-
tice comparison open

Theorem 7
 B^* has VCG-equivalent
payoffs and LEF/stability
consequences under explicit
ranked/canonical-tail
certificate hypotheses

Theorem 8
generalized-English auction
has unique PBE with finite-
 B^* threshold strategy and
VCG outcome, conditional
on source-game PBE and
exact/no-overshoot
history obligations

Initial source inventory. The refreshed NBER Working Paper 11765 text exposes Remarks 1–3, Definition 4, Lemmas 5–6, and Theorems 7–8. This DAG intentionally keeps implementation certificate layers out of the main graph. The detailed Lean endpoints, finite schedule wrappers, and no-overshoot review interfaces are recorded in the README, PaperInterface, post-paper audit report, and Theorem 8 source proof plan.

Current Theorem 8 boundary. The finite payoff algebra, threshold strategy behavior, terminal-record formulas, completed-rank and all-rank no-overshoot endpoints, shifted continuation-threshold bridge, and source-completion certificate plumbing compile. Belief-explicit state-game projections and cold-start sorted-schedule completion endpoints now route through source-layer bridge theorems. The full paper theorem remains conditional until the real generalized-English source game supplies the arbitrary-history generation and PBE identification needed to remove the remaining source-completion boundary.